

SOME INFORMAL AND NON-RIGOROUS INTUITION ABOUT THE COMPLEMENTARY SLACKNESS CONDITION

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The framework is the following: you aim to minimize a convex functional $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ under the constraints $\forall j \in \llbracket 1, q \rrbracket, h_j(x) \leq 0$. The Lagrangian associated to this problem is

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^q \lambda^{(j)} h_j(x).$$

Assume that the Slater's condition holds. Then, for x^* to be a solution to the problem, the following conditions should be satisfied:

- **Stationarity:** $\nabla f(x^*) + \sum_{j=1}^q \lambda^{(j)} \nabla h_j(x^*) = 0$;
- **Primal feasibility:** $\forall j, h_j(x^*) \leq 0$;
- **Dual feasibility:** $\forall j, \lambda^{(j)} \geq 0$;
- **Complementary slackness:** $\forall j, \lambda^{(j)} h_j(x^*) = 0$.

To get some intuition about the last condition, let assume that there is only one inequality constraint (i.e. $q = 1$). The feasible set of the problem is $C = \{x \in \mathbb{R}^d : h(x) \leq 0\}$. Its boundary is $\partial C = \{x \in \mathbb{R}^d : h(x) = 0\}$ (see figure 0.1).

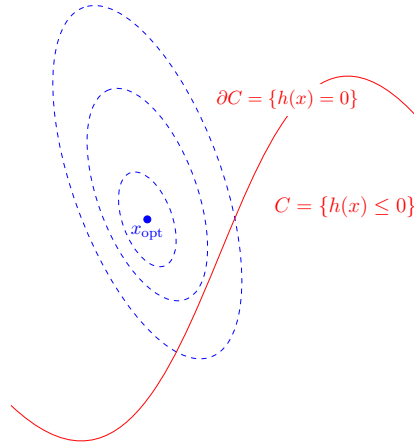


FIGURE 0.1. Visualization always helps (x_{opt} is the global minimizer of f , *without* constraints)

The idea is the following: if you want to minimize f , you'll essentially want to follow $-\nabla f$ (i.e. move towards x_{opt}), since it is the direction along which f decreases the fastest. However, doing so will eventually make you leave the feasible region C , which is not what you want. To avoid this, you want something that prevents you to cross the boundary ∂C when following $-\nabla f$. This is *exactly* what the Lagrange multiplier λ do:

- When $h(x) < 0$ (the constraint is inactive), you're inside the feasible region, so you can still move in the direction of $-\nabla f$. Hence, the stationarity condition reads $\nabla f(x) = 0$ (this is your goal), *id est* $\lambda = 0$.
- However, when $h(x) = 0$ (the constraint is active), you're at the boundary of the feasible region. Thus, you cannot follow $-\nabla f$ without leaving that region. Hence, you need a "force" that retains you from escaping C . This force is λ : when $\lambda > 0$, what you aim to solve is $\nabla f(x) + \lambda \nabla h(x) = 0$ (this is again the stationarity condition).

To summarize, the complementary slackness encodes the fact that when a constraint is inactive, it should not participate in the optimality condition. Both above cases are summarized by $\lambda h(x) = 0$.

If you have any questions, do not hesitate to contact me!